THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS UNIT TEST SET C

Chapters 11–14: **Exploring Some Geometric Themes, Tales by Dots and Lines, Algebra Play, Area** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. The number of edges of a pentagonal prism is: 1

(a) 14 **(b)** 18 **(c)** 16 **(d)** 15

2. In the Sierpinski triangle, the number of small shaded triangles after stage 6 is: 1

(a) 729 **(b)** 732 **(c)** 1458 **(d)** 728

3. A cylinder has how many curved surfaces? 1

(a) 3 **(b)** 1 **(c)** 2 **(d)** 0

4. The median of an even number of observations is: 1

(a) mean of the two middle values **(b)** the larger middle value
(c) the mean **(d)** the first value

5. The mean is called the 'balancing point' because: 1

(a) the total deviation on both sides is equal
(b) it is the most frequent
(c) it is the middle value
(d) it is the largest

6. The mean of the data 12, 15, 18, 20, 25 is: 1

(a) 20 **(b)** 16 **(c)** 19 **(d)** 18

7. The algebraic expression for 'five more than twice a number x' is: 1

(a) $x + 10$ **(b)** $2(x + 5)$ **(c)** $2x + 5$ **(d)** $5x + 2$

8. Think of a number n. Add 8, multiply by 3, then subtract 24. The result is: 1

(a) $3n + 8$ **(b)** $8n$ **(c)** $3n$ **(d)** $n + 8$

9. The two-digit number with tens digit 6 and ones digit 4 can be written as: 1

(a) $10 \times 6 + 4$ **(b)** $6 + 4$ **(c)** 6×4 **(d)** $10 \times 4 + 6$

10. The area of a triangle with base 14 cm and height 9 cm is: 1

(a) 126 cm² **(b)** 23 cm² **(c)** 63 cm² **(d)** 77 cm²

11. 1 m² equals: 1

(a) 10000 cm² **(b)** 1000 cm² **(c)** 100000 cm² **(d)** 100 cm²

12. A rectangle and a parallelogram on the same base and between the same parallels have: 1

(a) equal area **(b)** different area **(c)** double area **(d)** half area

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 10–15 words each.

13. How many faces does a square pyramid have? Name their shapes. 2

14. Find the range of 14, 8, 22, 5, 19. 2

15. Show that the difference of squares of consecutive numbers is odd. 2

16. Why is perimeter not a measure of area? Give an example. 2

17. Find the base of a parallelogram of area 108 cm² and height 9 cm. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 30–40 words each.

18. A Sierpinski triangle starts with one shaded triangle of side 1 unit. After stage 6, find the number of shaded triangles and the side of each. 3

19. Find the mean and the median of: 30, 35, 40, 45, 50, 60, 65. Which is greater? 3

20. A 2×2 block of a calendar has top-left date 4. Find the four dates, their sum, and show sum $= 4k + 16$. 3

21. A rectangular plot is 27 m by 6 m. Find its area and the cost of fencing at ₹ 30 per metre. 3

22. The mean of 4 observations is 11. If 3 of them are 8, 10, 14, find the remaining observation. 3

OR

The data 45, 52, 38, 60, 55, 40 is given. Compute the mean and median and comment on which is a better measure. 3

23. Use a number pyramid with bottom row 2, 9, 4: find all rows and the top, and write the top in terms of the bottom numbers. 3

OR

Two numbers add to 37. Find the numbers that give the largest product and the product. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

24. **Case study:** The monthly rainfall (cm) in a town is Jan 2, Feb 3, Mar 5, Apr 8, May 12, Jun 20. 5

(i) Draw (describe) the line graph.
(ii) Find the mean.
(iii) Between which months is the rise greatest?

OR

(i) The mean of 10 numbers is 24. If two numbers 18 and 30 are removed, find the new mean. 5
(ii) Explain.

25. (i) A rectangle's length is increased by 20% and its breadth decreased by 20%. Find the percent change in area. 5
(ii) Explain.

OR

(i) Derive the area of a triangle from a rectangle. 5
(ii) Find the area of a triangle with sides 13, 14, 15 using height 12 on side 14.

END OF QUESTION PAPER — Total: 50 Marks